Code & Results Unfrozen Encoders

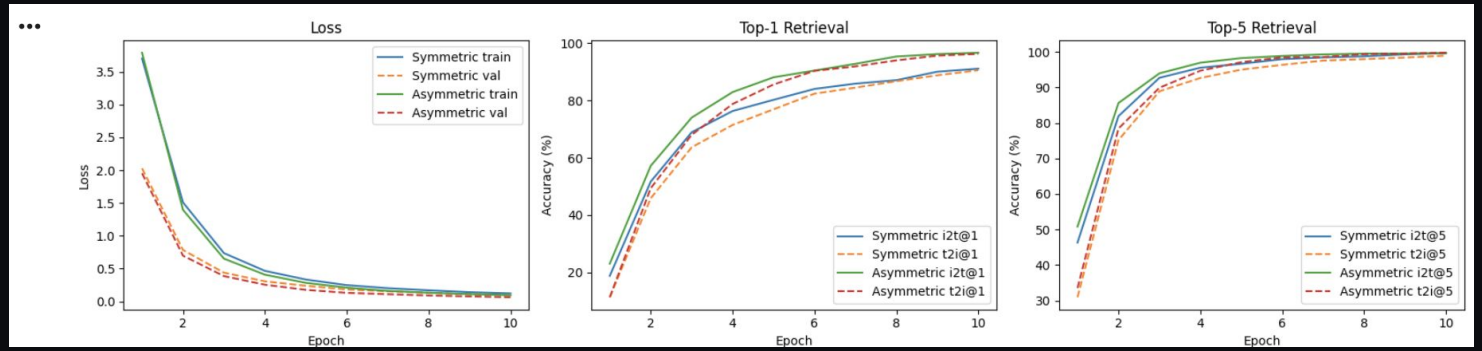
Aditya Katre

Link to Notebook

Link to presentation

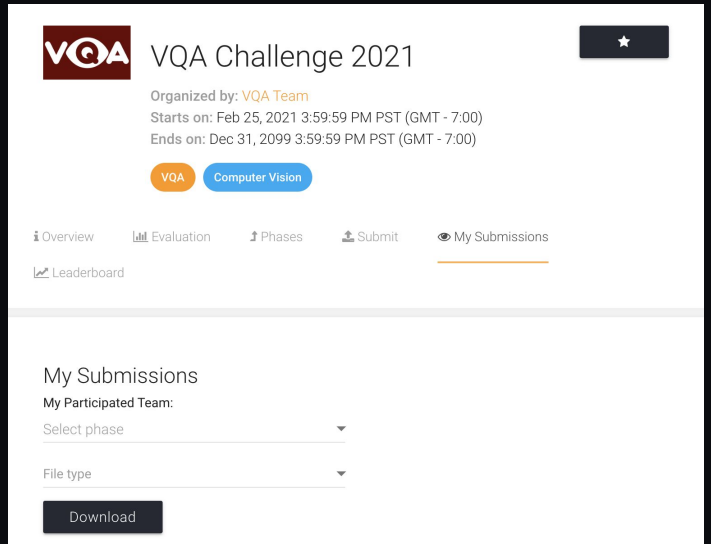
Changes Made to Notebook before Frozen Run

- Used L4 gpu instead of T4. L4 has more VRAM (needed for unfrozen)
- Updated graphs to conform to nandana's code. now looks similar to:



Changes Made to Notebook before Frozen Run contd.

- now exports test_predictions.json based on test dataset to upload to EvalAI
- predictions made based on test2015 image dataset folder



Changes Made to Notebook before Frozen Run contd.

- Set `FREEZE_ENCODERS=False`
- Implemented a resume/restore function based on saved .pt file:
- automatically deletes old epoch saves with lower accuracy to save storage

```
# --- RESUME LOGIC ---
if resume_from is not None:
    checkpoint_path = Path(resume_from)
    if checkpoint_path.exists():
        print(f"Resuming from checkpoint: {checkpoint_path}")
        checkpoint = torch.load(checkpoint_path, map_location=device)

        model.load_state_dict(checkpoint["model_state_dict"])
        optimizer.load_state_dict(checkpoint["optimizer_state_dict"])
        if scheduler is not None and checkpoint.get("scheduler_state_dict") is not Non
            scheduler.load_state_dict(checkpoint["scheduler_state_dict"])
        start_epoch = checkpoint["epoch"] + 1

        # Load previous history to keep appending
        history_file = METRICS_DIR / f"{run_name}_history.json"
        if history_file.exists():
            with open(history_file, "r") as f:
                history = json.load(f)

        if history:
            best_entry = max(history, key=lambda h: h.get("val_vqa_acc", 0))
            best_val_acc = best_entry.get("val_vqa_acc", 0)
            best_acc_epoch = best_entry["epoch"]
    else:
        print(f"Checkpoint not found at {checkpoint_path}. Starting from scratch.")
```

Changes Made to Notebook before Frozen Run contd.

- Hyperparameter changes:

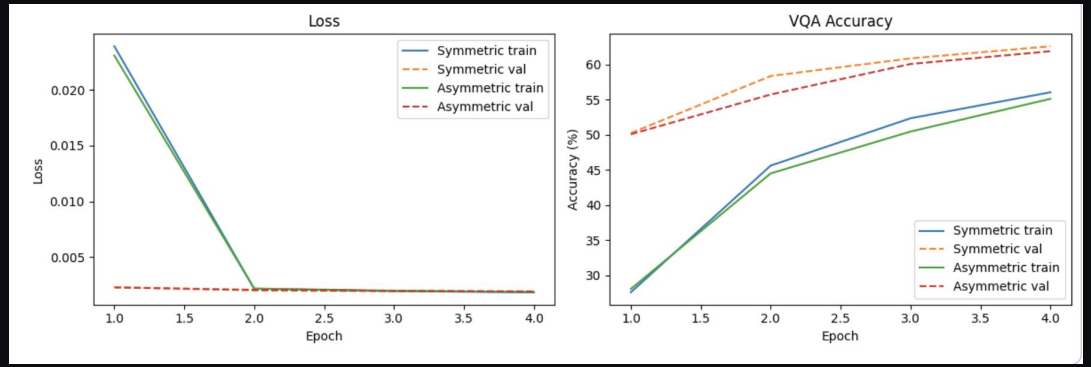
```
# Training
if FREEZE_ENCODERS:
    BATCH_SIZE    = 128    # precomputed features are small
    LEARNING_RATE = 1e-4
else:
    BATCH_SIZE    = 24     # backprop through ViT+RoBERTa needs ~10x more VRAM
    LEARNING_RATE = 1e-4   # fusion + classifier LR
    ENCODER_LR    = 1e-5   # pretrained encoder LR (10x smaller)
    WARMUP_EPOCHS = 1      # linear warmup to avoid catastrophic forgetting
```

- BATCH_SIZE=24 to fit VRAM constraints. During training VRAM was not limited; can be increased in future run
- LEARNING_RATE stayed the same
- ENCODER_LR=1*10^-5 sets learning rate for RoBERTa and ViT aka. step size for optimization
- WARMUP_EPOCHS = 1 helps avoid catastrophic forgetting. gives stastics during gradient descent time to stabilize before updates are made.
    - avoids losing the preexisting weights in ViT and RoBERTa and overwriting with VQA

Training Results

- Took 14 hours over 4 epochs. used to verify that everything is working
- trained on full dataset for eventual submission to EvalAI

Symmetric	62.63
Asymmetric	61.92



NameError During Training

```python
# Generate attention maps for sample validation images
for i in range(min(5, len(raw_val_ds))):
    image, input_ids, attention_mask, answer_target = raw_val_ds[i]
    question = raw_val_ds.samples[i]["question"]
    tokens = decode_tokens(input_ids)

    attn = get_attention_weights(e2e_asymmetric, image, input_ids, attention_mask)

    fig = plot_image_attention(attn["txt_to_img"], image, tokens, question)
    fig.savefig(FIGURES_DIR / f"attn_img_{i}.png", dpi=150, bbox_inches="tight")
    plt.show()

    fig = plot_text_attention(attn["img_to_txt"], tokens, question)
    fig.savefig(FIGURES_DIR / f"attn_txt_{i}.png", dpi=150, bbox_inches="tight")
    plt.show()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_6424/2713141154.py in <cell line: 0>()
      5     tokens = decode_tokens(input_ids)
      6
----> 7     attn = get_attention_weights(e2e_asymmetric, image, input_ids, attention_mask)
      8
      9     fig = plot_image_attention(attn["txt_to_img"], image, tokens, question)

NameError: name 'e2e_asymmetric' is not defined
```

Next steps: Explain error

- caused by heat maps/attention referring to hard coded frozen encoder instead of the appropriate frozen code.
- fixed by simply changing one line.

- resulted in a loss of completed symmetric epoch 4 as a result
  - successfully saved all symmetric epochs and the first three symmetric epochs

Next Steps

1. error has been fixed (one line change)
2. load the saved .pt files and upload them to test set evaluation
   a. determine whether our predictions conform to their expected formatting

3. If after completing #1-2 (making changes to code as needed) complete run over 10-15 epochs
   a. plenty of colab credits left from edu plan
4. Generate json files based on unfrozen notebook that will be submitted to EvalAI
5. continue adding to overleaf doc based on last week's comments